MAHARAJA AGRASEN UNIVERSITY

End Term Examination

Semester II, May 2025

Course: BCA
Subject: Mathematics II
Subject Code: BCA-202
Time: 3 hrs.
Max. Marks: 60

Note:- Attempt Five Questions in all.
Part A is compulsory. Attempt any 10 parts from Q1.
Select One Question from Each Unit of Part B.

PART A

Q1. **(i)** Define a set with two example.

(ii) Define subset of a set with two examples.

(iii) Define Cardinality of a set with an example.

(iv) If $A = \{1,3,5,7,9\}$ and $B = \{2,3,5,7\}$. Then find $A \cup B$ and $A \cap B$.

(v) If $\begin{bmatrix} 2 & 3 \\ 7 & 5 \end{bmatrix} + \begin{bmatrix} 2 & x-2 \\ y-1 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 7 & 10 \end{bmatrix}$, then find x and y.

(vi) Find the value of x such that $A = \begin{bmatrix} 2 & x+2 \\ 1 & 2 \end{bmatrix}$ is a singular Matrix.

(vii) Evaluate $\frac{d}{dx}(x^2+1)$ and $\frac{d}{dx}(\cot x)$.

(viii) Evaluate derivative of $\sin(10x^2)$ with respect to x

(ix) Evaluate $\lim_{x \to 1} \frac{x-1}{x^9-1}$.

(x) Evaluate Left hand and Right hand limit of the function $f(x) = 4x+5$ at $x = 0$.

(xi) Evaluate derivative of $e^{(3x+2)}$ with respect to x.

(xii) If $A = \{1,2,3\}$, then write $P(A)$. **(2 x 10 = 20)**

PART B

UNIT I

Q2. **(a)** Define Cartesian product of a set with an example.

(b) Define trigonometric, logarithmic and exponential function with examples of each.

(c) $A = \{1,3,5\}$ and $B = \{2,3,5\}$. Then find $A \times B$ and $B \times A$. **(10)**

Q3. **(a)** Define function with an example.

(b) Define relation on a set with an example.

(c) If $A = \{4,6,8,10,12\}$, $B = \{3,6,9,12,15,18\}$ and $C = \{1,2,3,4,5,6,7,8,9,10\}$. Then find $A \cup B \cup C$ and $A \cap B \cap C$. **(10)**

UNIT II

Q4. Find the determinant of the following Matrices

(i) $A=\begin{bmatrix}5&0&0\\0&2&0\\0&0&3\end{bmatrix}$ (ii) $A=\begin{bmatrix}2&0&-1\\0&1&5\\1&3&-4\end{bmatrix}$ **(10)**

Q5. Find the inverse of the following Matrices $A=\begin{bmatrix}-3&7&-5\\4&-1&12\\-2&9&-1\end{bmatrix}$. **(10)**

UNIT III

Q6. Discuss the continuity of the functions

(a) $f(x)=\begin{cases}\dfrac{|x-2|}{x-2}, & \text{when } x=2\\ 1, & \text{when } x=2\end{cases}$ at $x=2$, (b) $f(x)=\begin{cases}1+x, & \text{when } x\le 2\\ 5-x, & \text{when } x>2\end{cases}$ at $x=2$, 1 3

(c) $f(x)=\begin{cases}2+x, & x\le 3\\ 8-x, & x>3\end{cases}$ at $x=3$. 1 5 (10)

Q7. Define three types of discontinuities for a function $f(x)$ at $x=a$ with examples. (10)

UNIT IV

Q8. Evaluate the derivative of the following functions with respect to x

(a) $\left(x^2+6x+8\right)^7$ (b) $\sin^2 x$ (c) xe^x (d) $x\log x$ (e) $\cos^2 x$ (10)

Q9. Find the maximum and minimum values of the function $2x^3-3x^2-36x+10$. (10)